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SUBJECT CODE NO: - RNP-8002
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group A & B]Mech, Civil, Comp. AI & ML,EEE
Examination January/February 2025
Engineering Physics

[Time: 3:00 Hours]**[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q. no's 2, 3, 4, 5, 6 and 7

SECTION A**Q1** Solve the following questions**20**

1. The energy of ultrasonic waves depends on:
 - a) Frequency of the wave
 - b) Amplitude of the wave
 - c) Both frequency and amplitude
 - d) Only the density of the medium
2. Which of the following devices is used to generate ultrasonic waves?
 - a) Loudspeaker
 - b) piezoelectric crystal
 - c) Microphone
 - d) Magnetron
3. The phenomenon where sound waves are absorbed by the medium and do not travel further is called-
 - a) Reflection
 - b) Diffraction
 - c) Absorption
 - d) Transmission
4. What is the primary component of an X-ray tube that generates X-rays?
 - a) Reflection
 - b) Diffraction
 - c) Absorption
 - d) Transmission
5. Ultrasonic waves have frequencies:
 - a) Below 20 Hz
 - b) Between 20 Hz and 20 kHz
 - c) Above 20 kHz
 - d) Between 1 Hz and 1 kHz
6. The accelerating potential for an X-ray is 40 kV. What should be the minimum wavelength of the X-ray?
 - a) 0.024 nm
 - b) 0.031 nm
 - c) 0.097 nm
 - d) 0.054 nm
7. Which of the following is ideal for the generation of laser pulse?
 - a) Ruby laser
 - b) Carbon dioxide laser
 - c) Helium neon laser
 - d) Nd-YAG laser

Q2	a) Write short notes. i. electromagnetic waves ii. electromagnetic spectrum b) Explain the formation of newton's ring?	05 05
Q3	a) Explain Bragg's Law? The wavelength of the X-rays is 0.071 nm which is diffracted by a plane of salt with 0.28 nm as the lattice constant. Determine the glancing angle for the second-order diffraction. Assume the value of the salt plane to be 110, and the given salt is rock salt. b) Explain time independent Schrodinger equation	05 05
Q4	a) Explain the Einstein's theory matter radiation interaction and A and B coefficient b) Write short notes. i. Diamagnetic material. ii. Paramagnetic material.	05 05
Q5	a) Explain He-Ne Leaser with diagram b) Explain the properties of leaser	05 05
Q6	a) Explain De-Broglie's concept of matter wave b) Explain p-p cycle & c-n cycle	05 05
Q7	a) Explain the diffraction of light b) State and explain Michelson interferometer	05 05